

Generative Artificial Intelligence in Qualitative Data Analysis: Analyzing—Or Just Chatting?

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SoDA 502 Reading Discussion

Paper: Nguyen & Welch (2026), *Organizational Research Methods*, 29(1), 3–39

Research Questions

- **Hype:** “30x faster” analysis
- **RQ1:** What kind of tool is an LLM?
- **RQ2:** Is it fit for qualitative analysis?
- **Thesis:** Not suited; serious epistemic risks

Theoretical Framing: Instruments in Science

- Tools shape knowledge (Alvarado 2023)
- Example: tape recorder → transcripts (Jones 2021)
- Four criteria for a valid tool:
 - Factual accuracy
 - Reliability
 - Transparency
 - Ethics

How LLMs Operate

- Predict the next word, one at a time
- Built to sound plausible, not to be true
- Different output on every run
- Black box: no audit trail
- Ethics: data leakage, energy use
- RAG and APIs do not fix this

The Claims and the Authors' Stance

- **Claims:**
 - Automation: LLMs as “autocoders”
 - Augmentation: LLMs help interpret
 - Anthropomorphism: LLM as colleague
- **Stance:** QDA is interpreting social meaning
- So fluent text is not analysis

- **Own test:** 14 documents, 49,444 words
- **Tool:** ChatGPT-4o, three prompting styles
- **Benchmark:** 5 published studies
- **Qual-AI:** NVivo and QUALSOFT tested in depth

Results I: Automated Coding Failure

- Summaries, not real codes
- Quotes not in the dataset
- Misleading theme labels
- Wrong page numbers

Results II: Iteration & Lost Efficiency

- Polished output only after 8 rounds
- Focused on parts, ignored the rest
- Constant manual checking
- Researcher becomes content moderator

Results III: Dedicated Qual-AI Tools

- Best on one transcript only
- Banal themes; key themes missed
- Arbitrary quote boundaries
- People and companies mixed up
- No audit trail

Five Epistemic Risks in QDA

- 1 **Category error:** prediction is not analysis
- 2 **Unreliable outputs**
- 3 **Anthropomorphic fallacy:** chatting is not dialogue
- 4 **Causal misattribution:** blaming the prompts
- 5 **Oracle effect:** errors explained away

The Oracle Effect: Rationalizing Errors

- Studies found errors, yet stayed positive
- Lee et al.: varied themes = “different angles”
- De Paoli: “Human-AI collaboration”
- Hitch: errors as “alternative analysis”
- Hazard: machine error is not human reflexivity

Critical Reflection on the Paper

Strengths

- Explains the technology first
- Triangulated evidence
- Clear criteria

Questions

- One topic, one main tester
- No error rates
- Inductive coding only
- Others read uncharitably?

Conclusion: Form vs. Substance

- Looks like analysis, is not
- Interpretation is human
- Risk: homogenized research
- **Chatting is not analyzing.**

Thank You!

Discussion Questions:

- 1 Can AI help with brainstorming without harming reflexivity?
- 2 Does the critique hold for codebook-based coding?
- 3 Are the authors fair to cautious users?
- 4 What evidence would validate an LLM tool?

Supplementary: Validation Criteria

Criterion	Requirement	AR-LLM
Factual accuracy	No introduced errors or distortions	Fabrication-prone
Reliability	Stable, dependable results	Non-deterministic
Transparency	Verifiable audit trail	Black box
Ethics	Privacy, IP, sustainability	Data leakage; energy use

Supplementary: Cross-Study Comparison

Study	Data	Prompting	Deficiencies
Nguyen & Welch (2026)	GRiDPad corpus (49k words)	Zero/Few-shot, CoT	Fabricated quotes; re-prompting
Lee et al. (2024)	Diabetes podcast	Few-shot	Errors; inconsistency
Hitch (2024)	Long COVID article	Zero-shot	Literal themes
Sen et al. (2023)	2 interview transcripts	Zero-shot	Different codes, same text
De Paoli (2024)	Gaming/teaching interviews	Zero-shot (API)	Overlooked text; hallucinated codes
Turobov et al. (2024)	63 UN documents	Custom GPT (CoT)	Descriptive; random

Supplementary: Commercial Qual-AI Offerings

Software	AI Features	Paper's Observations
ATLAS.ti	AI coding; chat with documents	Feature review only; OpenAI-based
MAXQDA	AI coding; chat with data	Feature review only; OpenAI-based
NVivo	Summaries; child codes	Tested; vendor disclaims accuracy
QUALSOFT	Auto themes; querying (RAG)	Tested; arbitrary quotes; mixed-up entities
Quirkos	<i>No GenAI features</i>	Prioritizes reflexivity & transparency